HEART: COVERING (PERICARDIUM) AND EXTERNAL FEATURES OF HEART

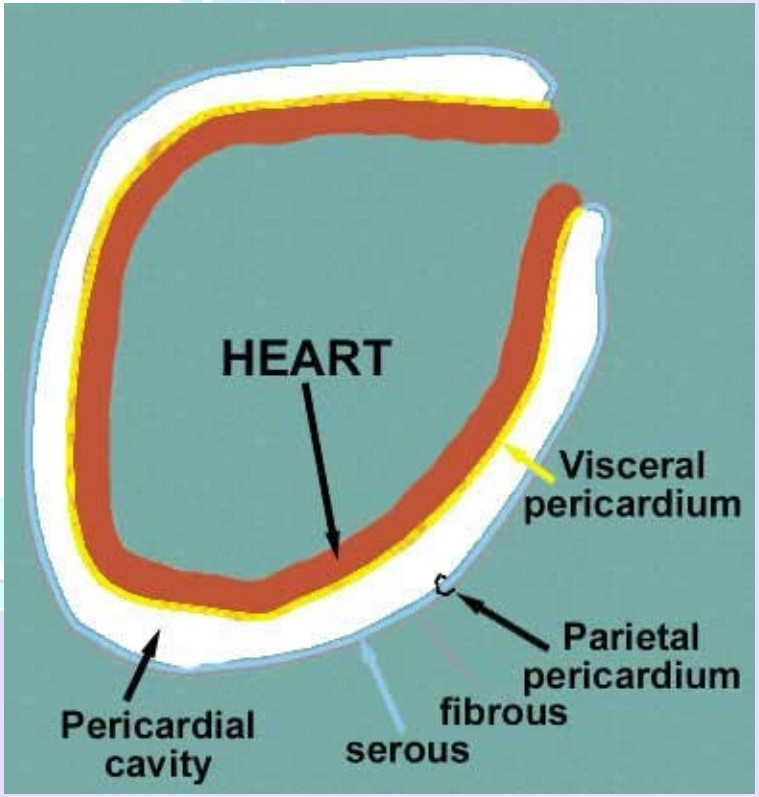
LEARNING OBJECTIVES

At the end of the demonstration, the student should be able to :

- **Define Pericardium**
- **Describe the layers of the heart walls and surrounding pericardium.**
- **Describe the anatomical position of the heart.**
- **Identify and describe the chambers and valves of the heart.**
- **Identify the major blood vessels to and from the heart.**
- **Describe the blood flow though the heart.**

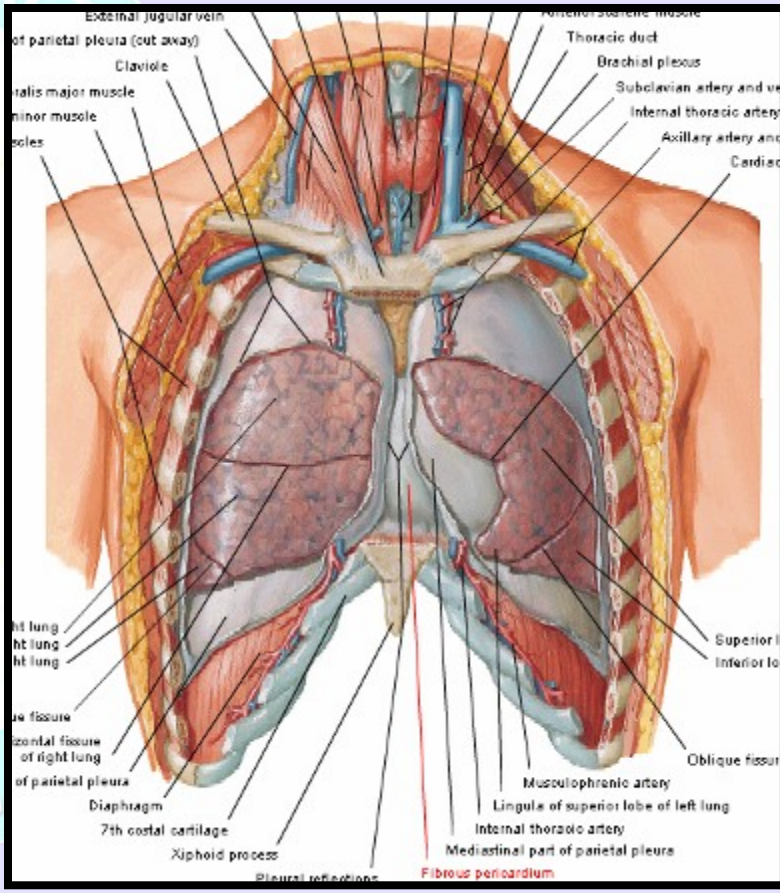
Definition

- **Pericardium:** The membranous sac filled with serous fluid that encloses the heart and the roots of the aorta and other large blood vessels.
  - Like other serous coverings, eg. Pleura & peritoneum it is also visceral and parietal layered
  - In addition there is an other covering FIBROUS PERICARDIUM

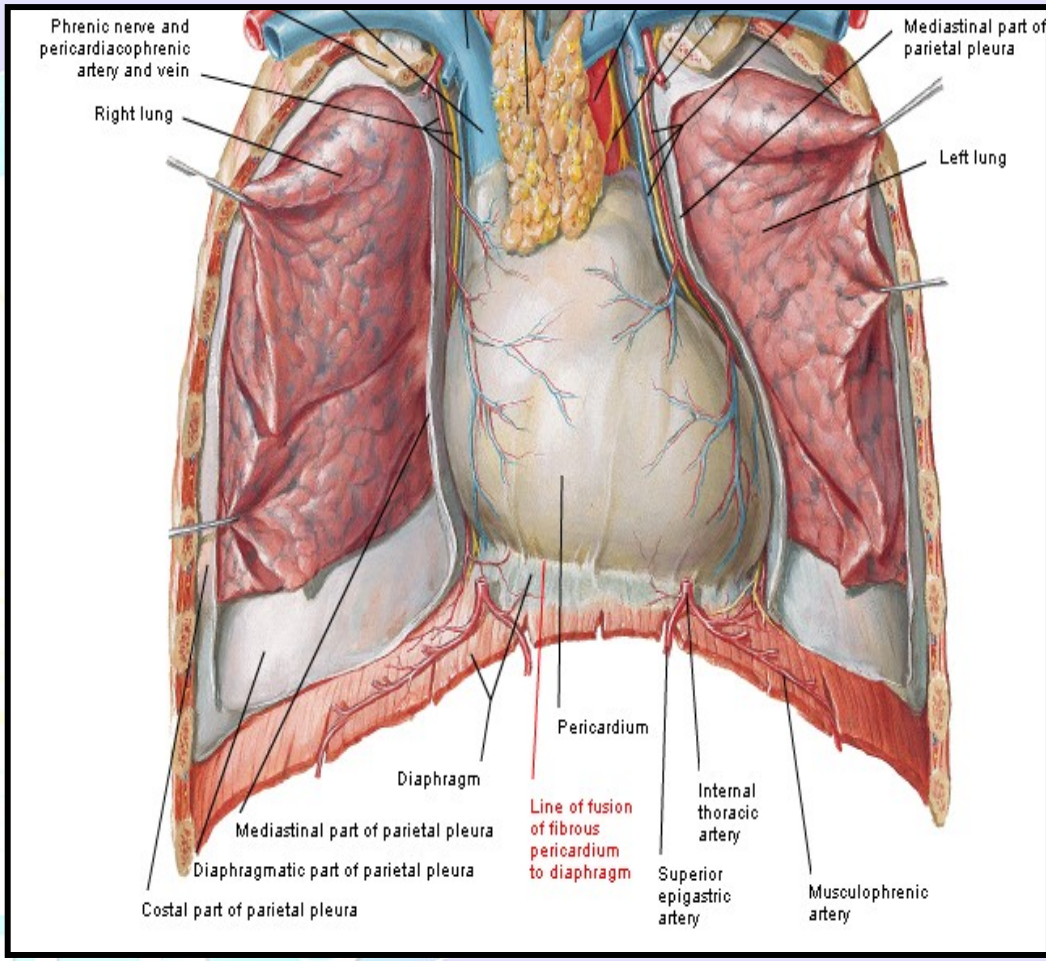


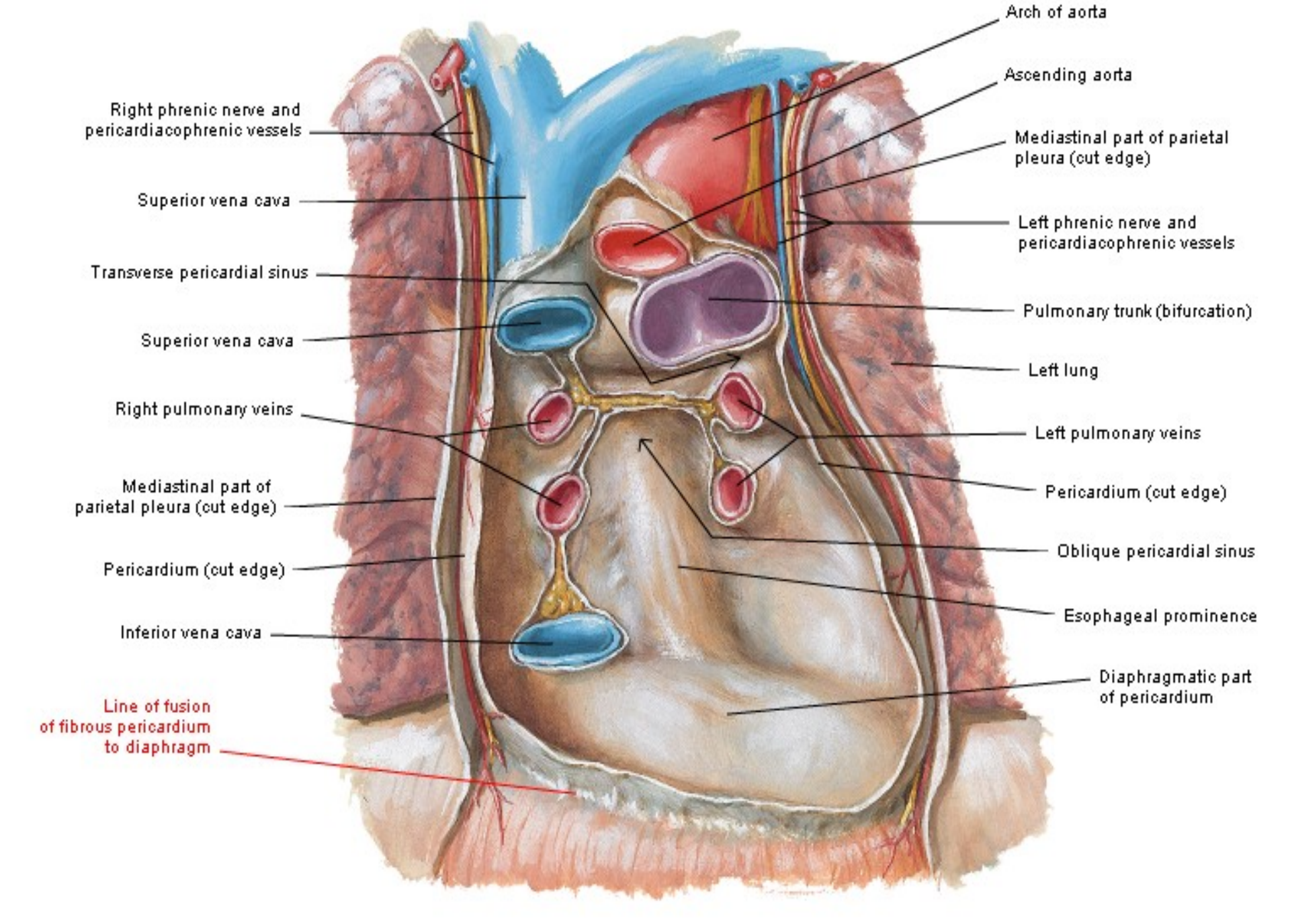
FIBROUS PERICARDIUM

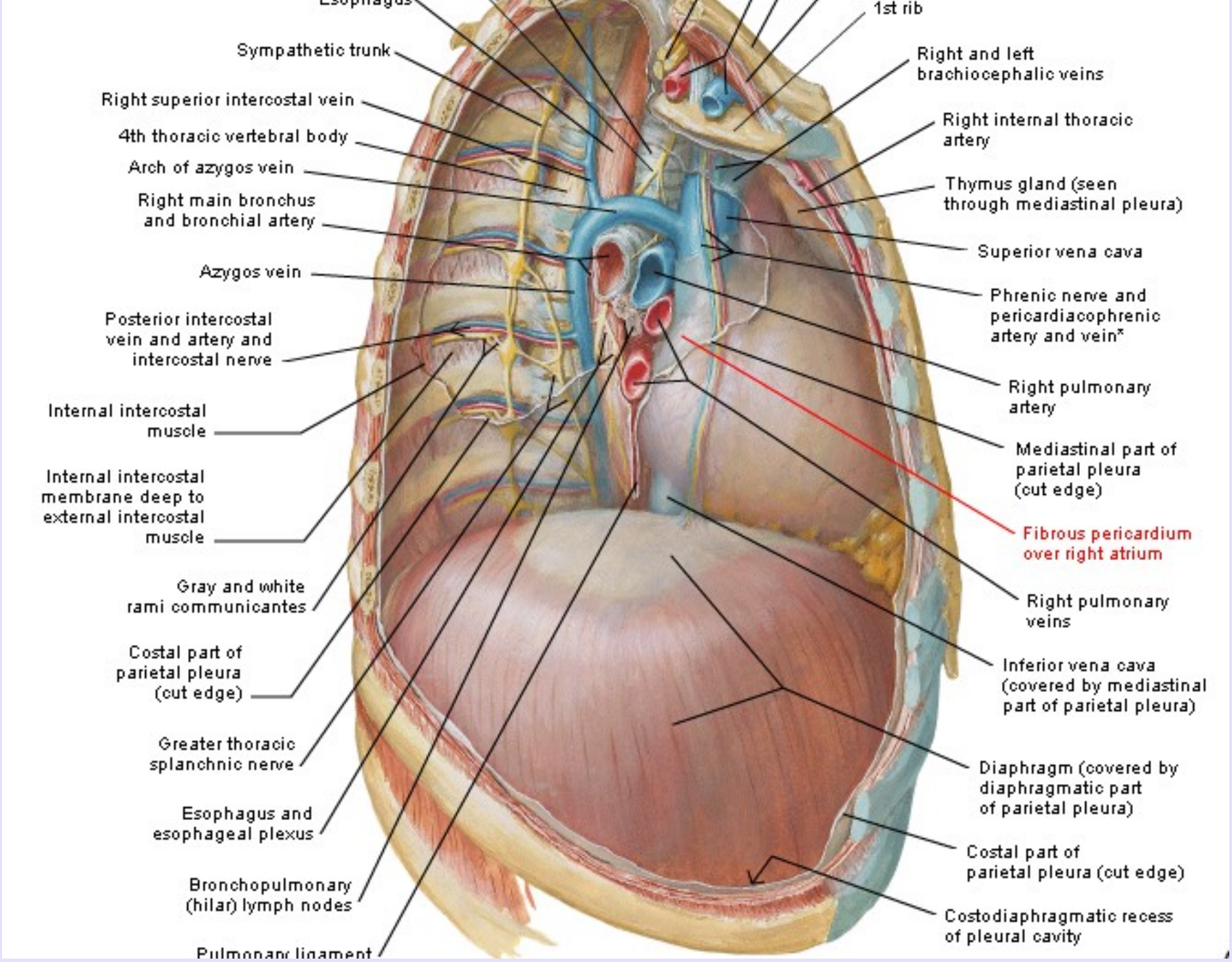
- It is a sac made up of connective tissue fully surrounding the heart with out being attached to it
- It is roughly conical in shape
- It is superiorly connected with tunica adventitia of great vessels
- Inferiorly it is connected with central tendon of diaphragm

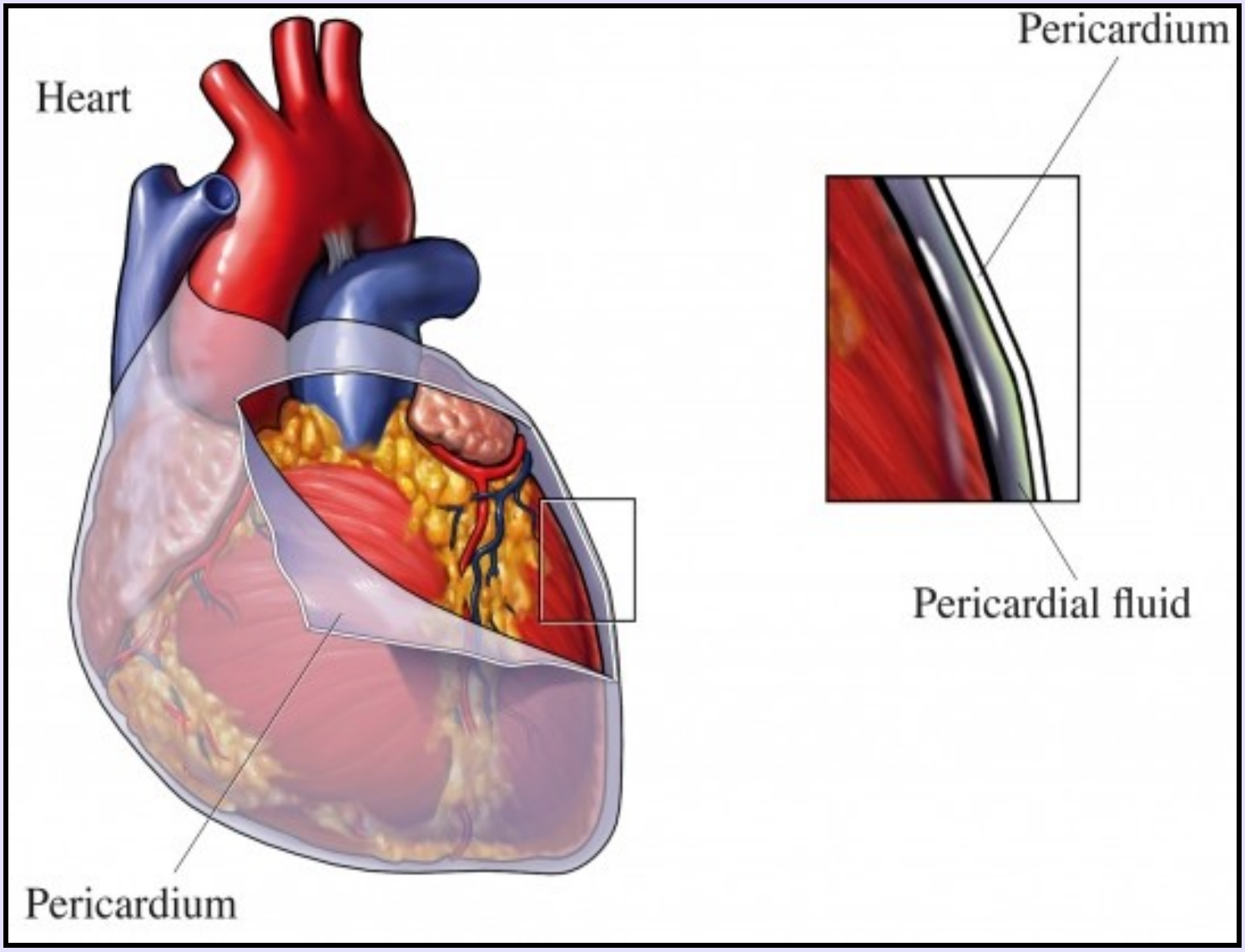


- **Anteriorly** it is separated from thoracic wall by lung & pleura, however some portion of it is in direct relation with left half of lower part of body of Sternum and left 4th &5th costal cartilages
- **Posteriorly** it is related to esophagus descending thoracic Aorta & posterior part of mediastinal surface of both lungs



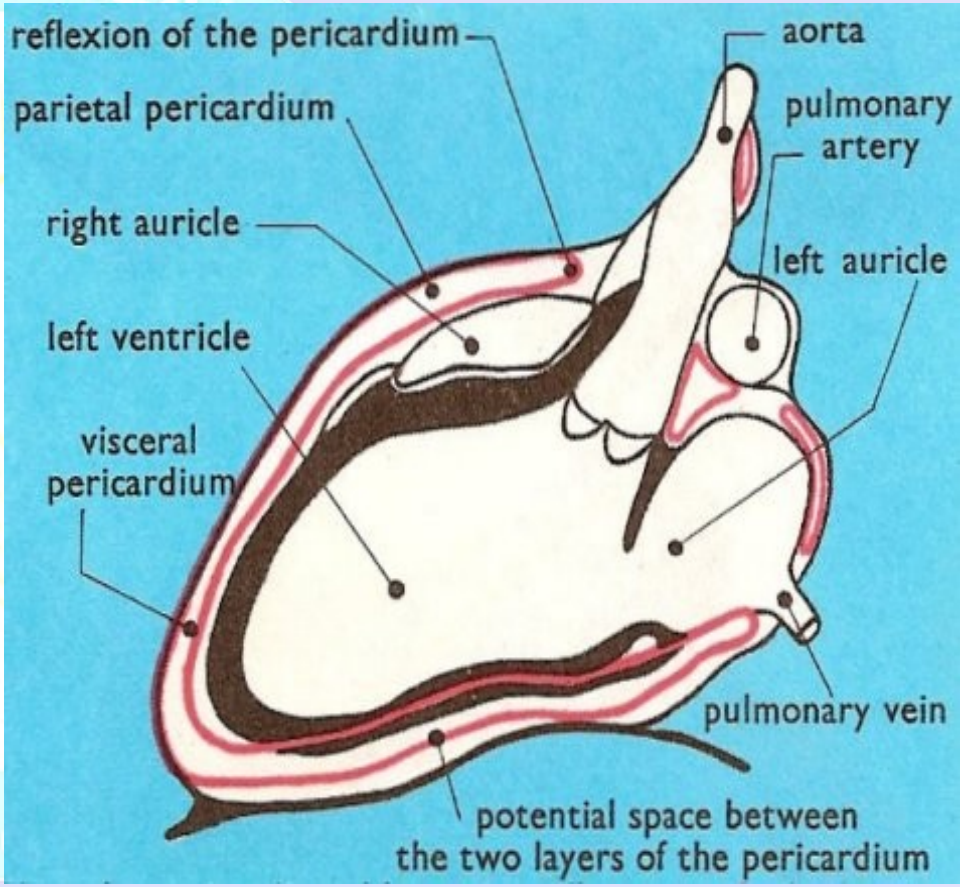

Arch of aorta
Ascending aorta
Right phrenic nerve and pericardiacophrenic vessels
Mediastinal part of parietal pleura (cut edge)
Superior vena cava
Left phrenic nerve and pericardiacophrenic vessels
Transverse pericardial sinus
Pulmonary trunk (bifurcation)
Superior vena cava
Left lung
Right pulmonary veins
Left pulmonary veins
Mediastinal part of parietal pleura (cut edge)
Pericardium (cut edge)
Pericardium (cut edge)
Oblique pericardial sinus
Esophageal prominence
Inferior vena cava
Diaphragmatic part of pericardium
Line of fusion of fibrous pericardium to diaphragm


Esophagus
1st rib
Sympathetic trunk
Right and left brachiocephalic veins
Right superior intercostal vein
Right internal thoracic artery
4th thoracic vertebral body
Thymus gland (seen through mediastinal pleura)
Arch of azygos vein
Right main bronchus and bronchial artery
Superior vena cava
Azygos vein
Phrenic nerve and pericardiacophrenic artery and vein*
Posterior intercostal vein and artery and intercostal nerve
Right pulmonary artery
Internal intercostal muscle
Mediastinal part of parietal pleura (cut edge)
Internal intercostal membrane deep to external intercostal muscle
Fibrous pericardium over right atrium
Gray and white rami communicantes
Right pulmonary veins
Costal part of parietal pleura (cut edge)
Inferior vena cava (covered by mediastinal part of parietal pleura)
Greater thoracic splanchnic nerve
Diaphragm (covered by diaphragmatic part of parietal pleura)
Esophagus and esophageal plexus
Costal part of parietal pleura (cut edge)
Bronchopulmonary (hilar) lymph nodes
Costodiaphragmatic recess of pleural cavity
Pulmonary ligament


Heart
Pericardium
Pericardium
Pericardial fluid

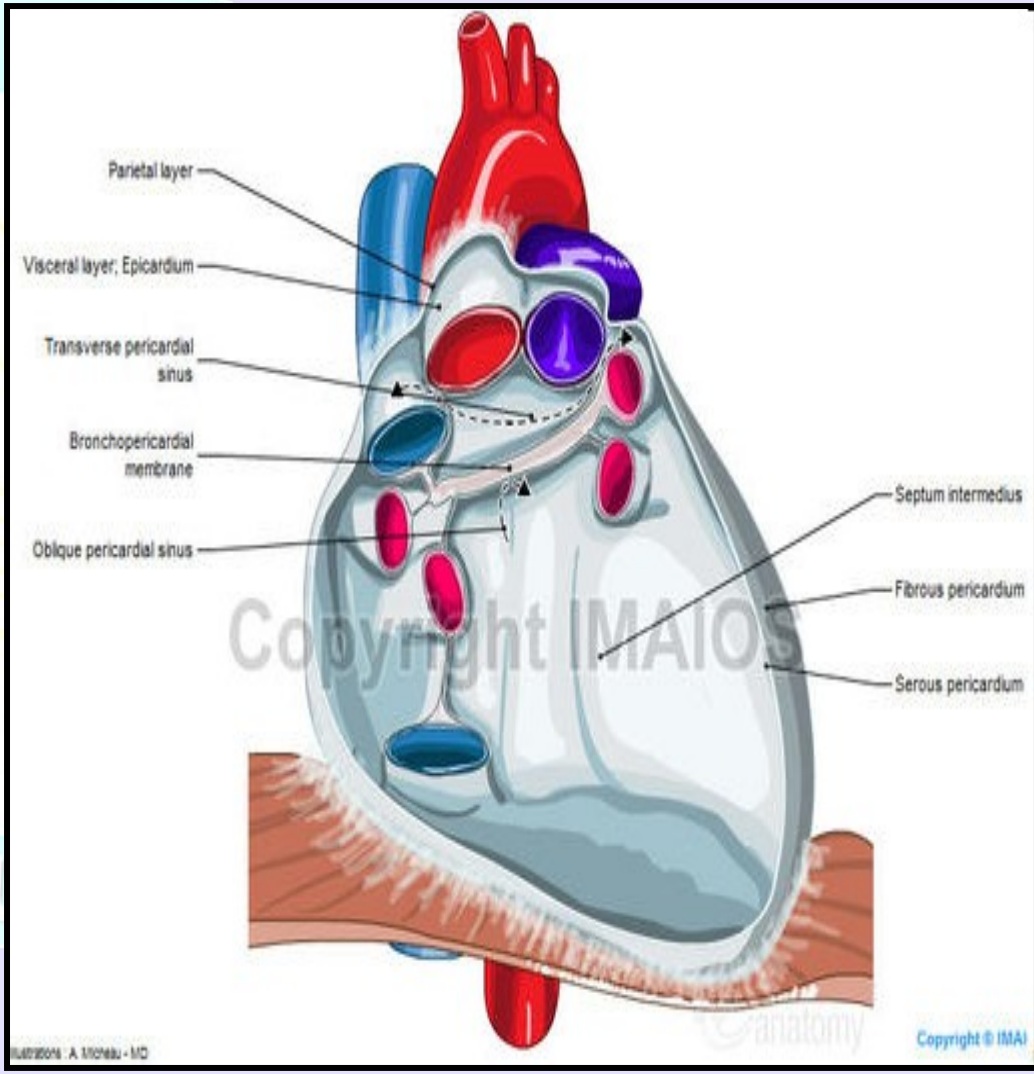
SEROUS PERICARDIUM

- It is closed sac within fibrous pericardium having Visceral & Parietal layer
- The visceral layer of serous pericardium (epicardium) covers the surface of the heart
- It also reflects onto the great vessels
- From around the great vessels, the serous pericardium reflects to line the internal aspect of the fibrous pericardium as the parietal layer of serous pericardium



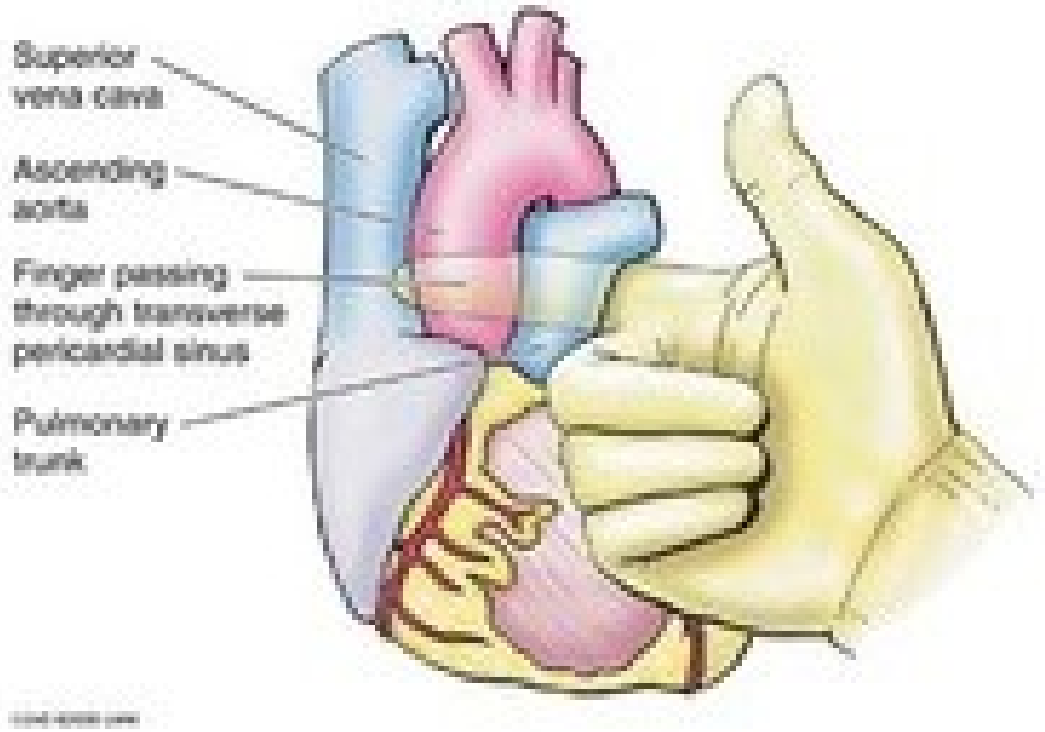
Transverse Sinus

- The transverse sinus is bounded anteriorly by the serous pericardium covering the posterior aspect of the pulmonary trunk and aorta, and posteriorly by the visceral pericardium covering the atria



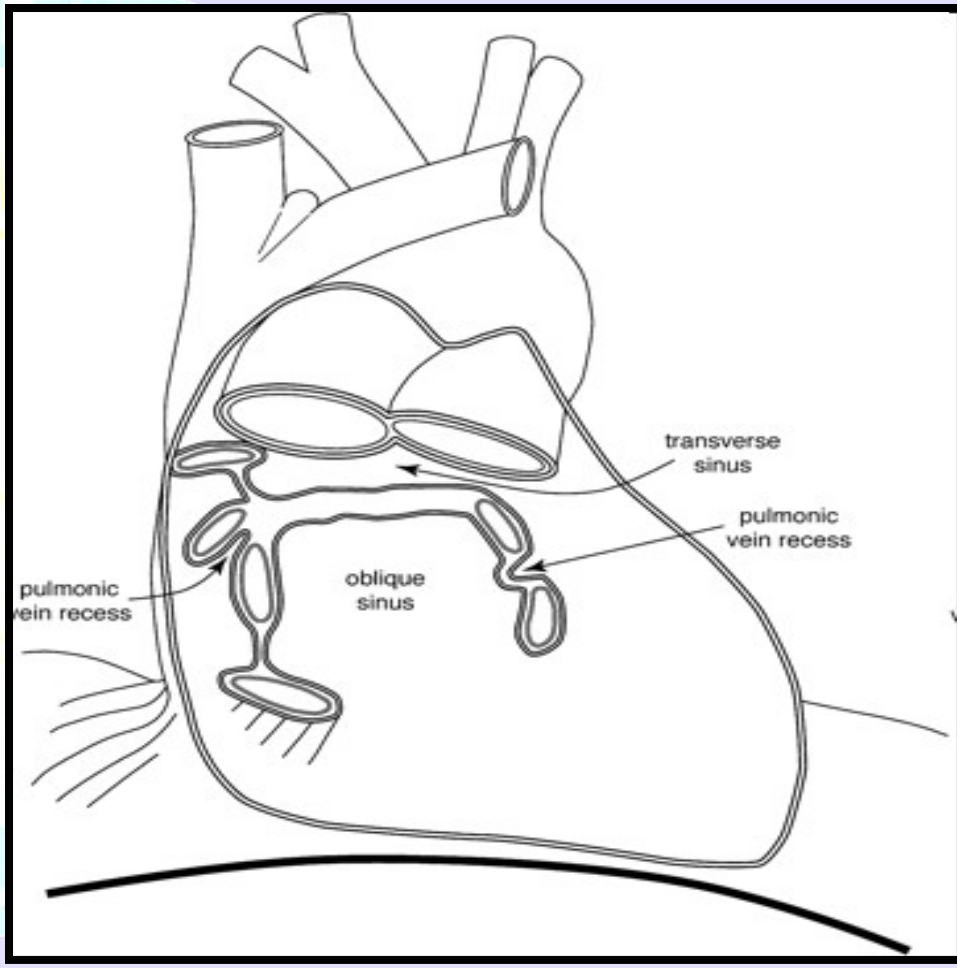
Transverse Sinus

- The transverse pericardial sinus is especially important to cardiac surgeons.
- After the pericardial sac has been opened anteriorly, a finger can be passed through the transverse pericardial sinus posterior to the aorta and pulmonary trunk.
- By passing a surgical clamp or placing a ligature around these vessels, inserting the tubes of a coronary bypass machine, and then tightening the ligature, surgeons can stop or divert the circulation of blood in these large arteries while performing cardiac surgery.


Superior vena cava
Ascending aorta
Finger passing through transverse pericardial sinus
Pulmonary trunk

OBLIQUE SINUS

- The oblique sinus is bounded
  a. anteriorly by the visceral layer of serous pericardium covering the left atrium
  b. posteriorly by the parietal layer of serous pericardium lining the fibrous pericardium,
  c. superiorly and laterally by the reflection of serous pericardium around the four pulmonary veins and the superior and inferior vena cavae



CARDIAC TAMPONADE

- Cardiac tamponade (heart compression) is due to critically increased volume of fluid outside the heart but inside the pericardial cavity; e.g., due to stab wounds or from perforation of a weakened area of the heart muscle after heart attack (hemopericardium).

The Heart.

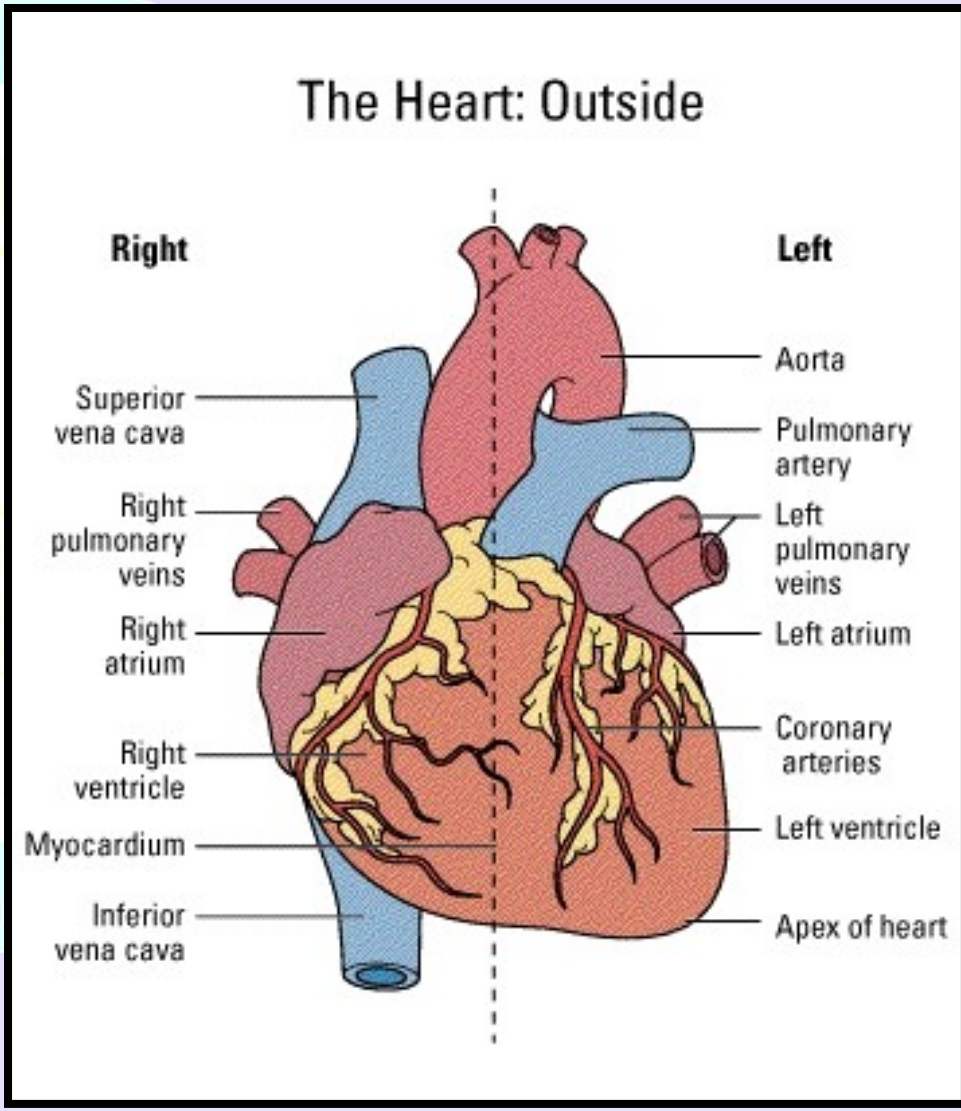
Position & External Features

POSITION

- The heart is located directly on **top of the diaphragm behind the sternum**.
- It is positioned in the **middle mediastinum, between the left and right lungs.**

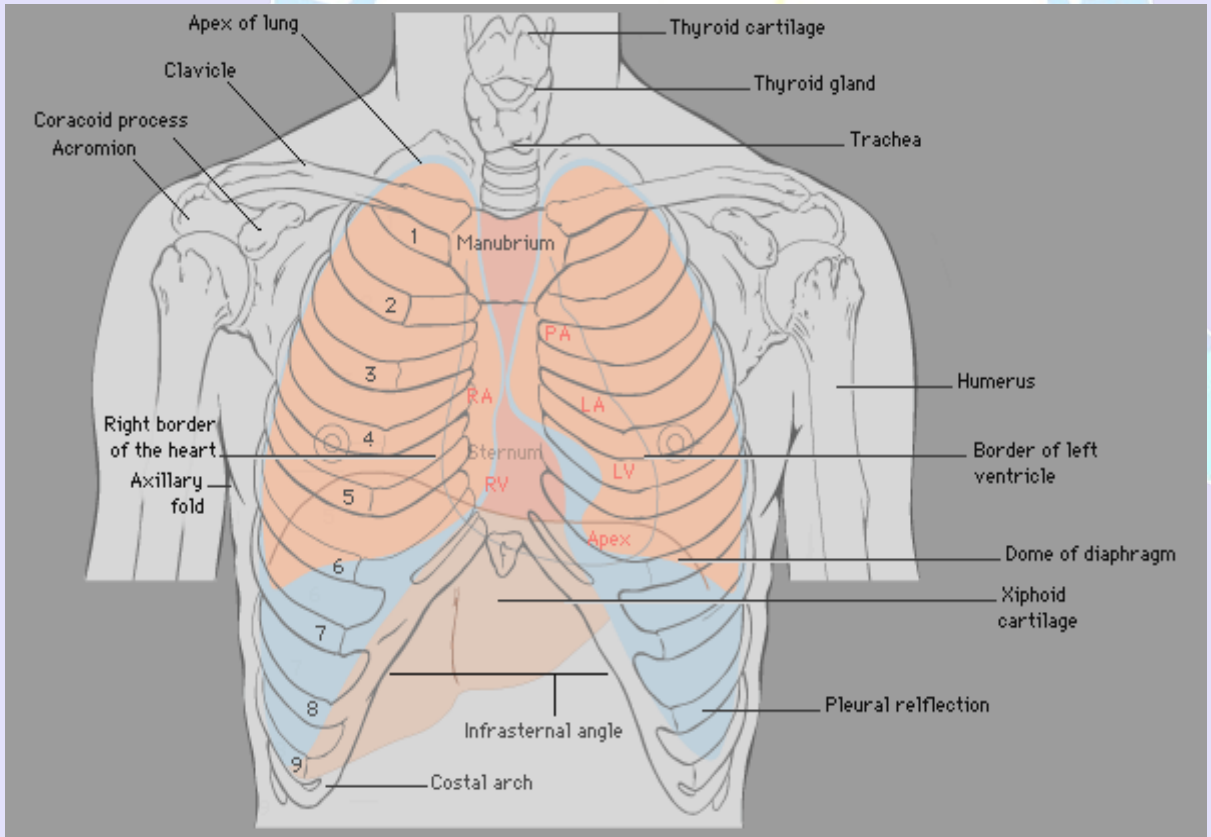
Structure of the Heart:

- The heart is a **myocardial muscular pump** consisting of **four chambers**, **two auricles**, **four valves** and a **muscular septum** all enclosed within a fluid filled sac, the **pericardium.**



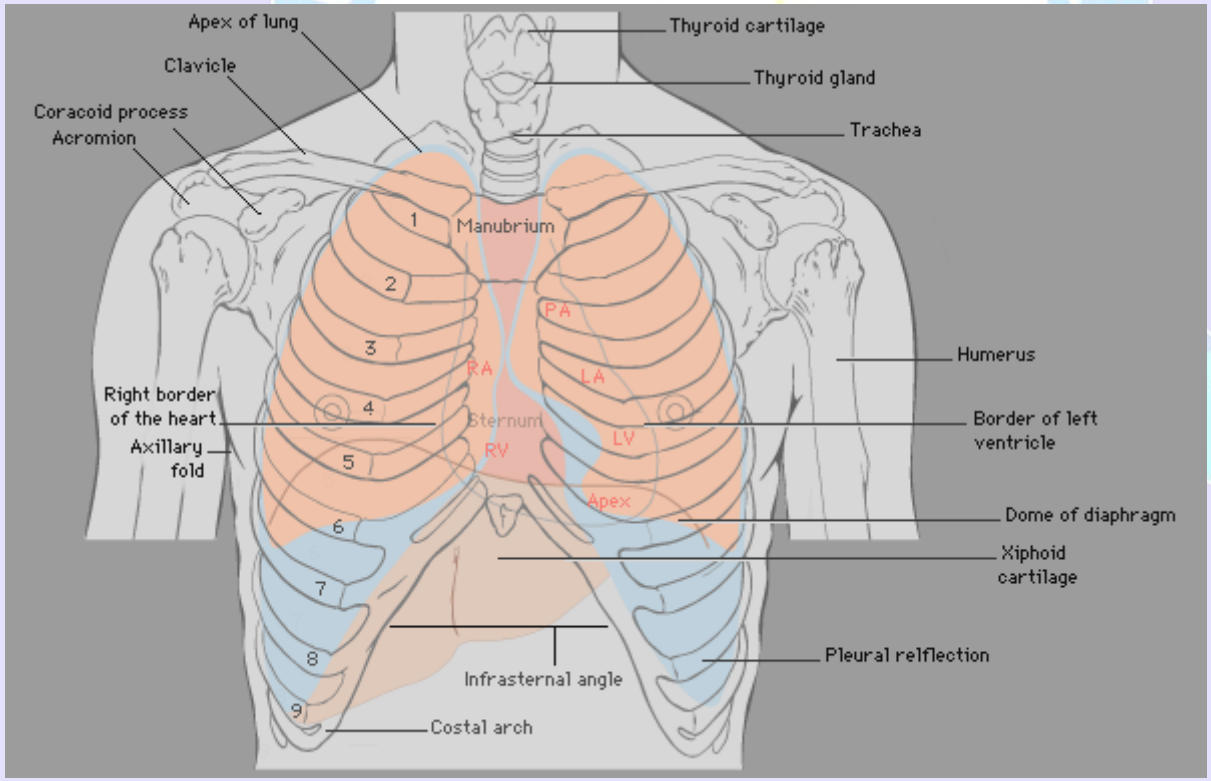
Position

- Right border consists entirely of the **right atrium**.
- Inferior border is made up mostly of **right ventricle** with a small portion of **left ventricle.**
- Left border is mostly **left ventricle** , **auricle of left atrium** forming uppermost part.



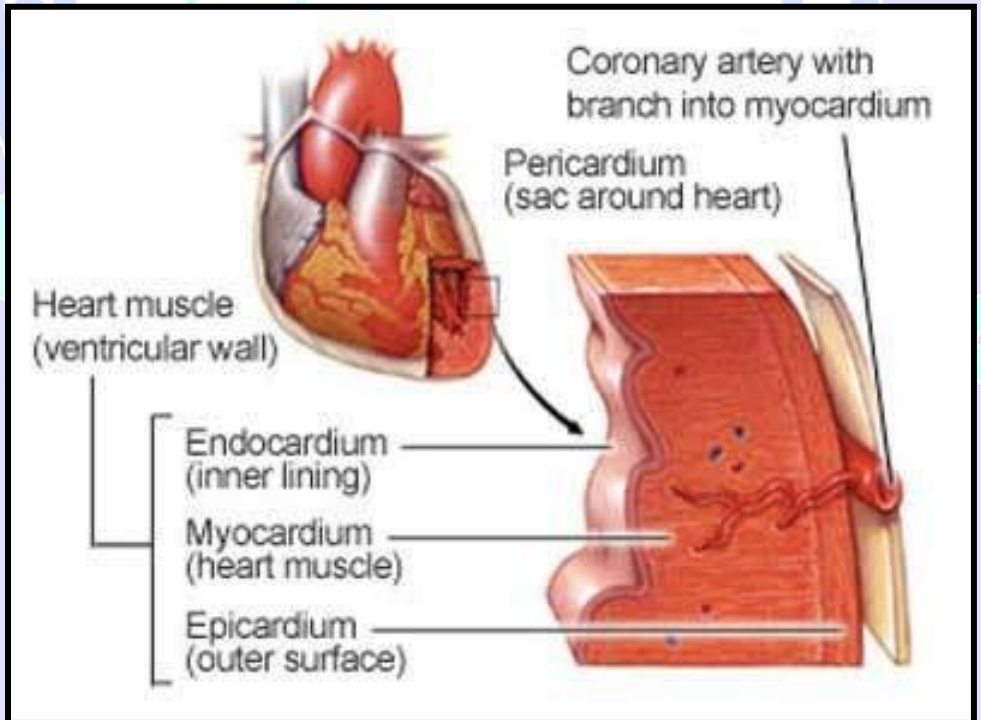
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Heart Wall

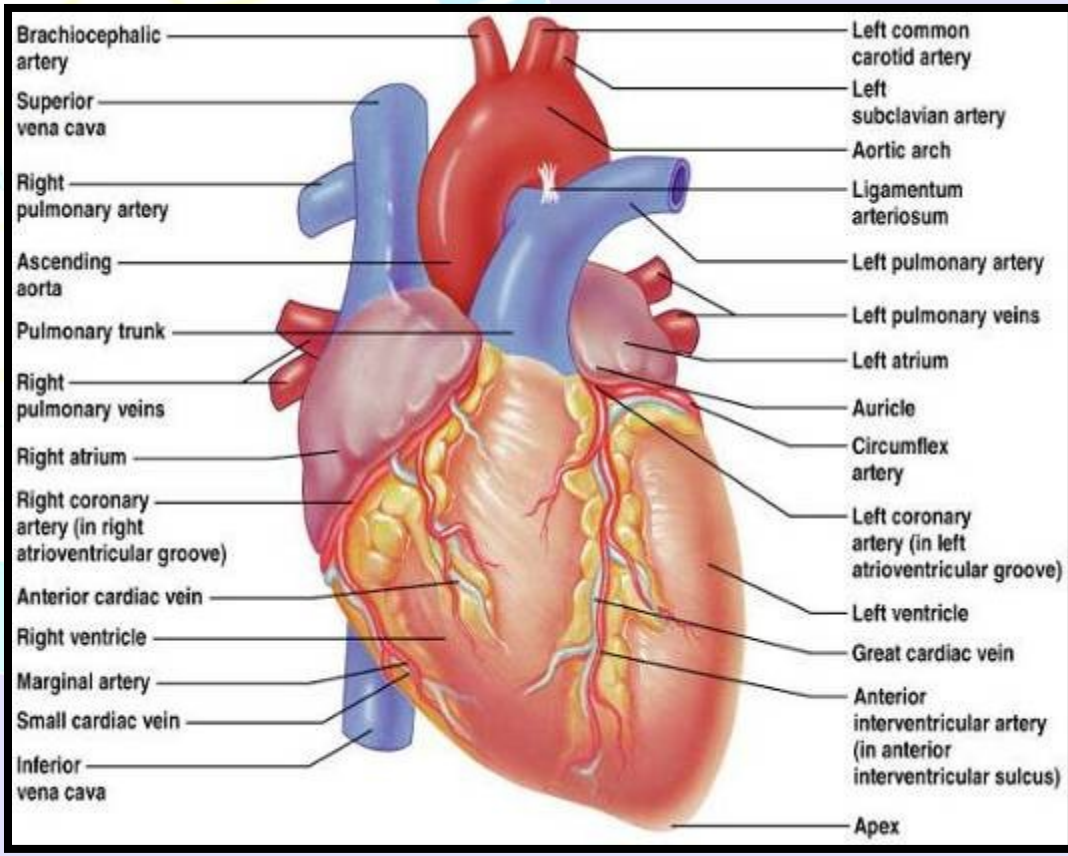
- **Epicardium** – visceral layer of the serous pericardium
- **Myocardium** – cardiac muscle layer forming the bulk of the heart
- **Fibrous skeleton of the heart** – crisscrossing, interlacing layer of connective tissue
- **Endocardium** – endothelial layer of the inner myocardial surface



External Heart: Major Vessels of the Heart (Anterior View)

- **Vessels** returning blood to the heart include:
  - **Superior** and **inferior venae cavae.**
  - Right and left **pulmonary veins**.

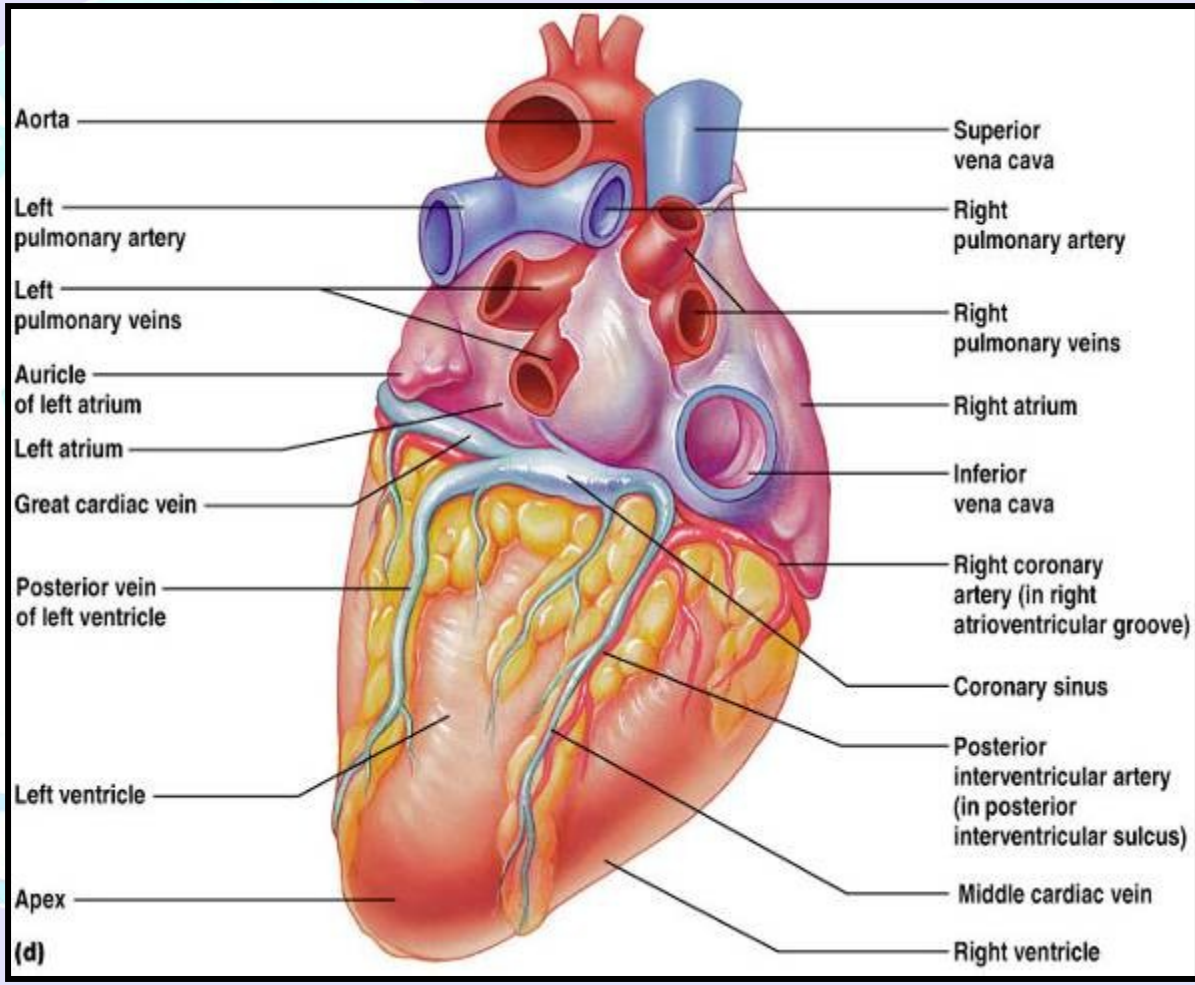
- **Vessels** conveying blood away from the heart include:
  - **Pulmonary trunk**, which splits into right and left pulmonary arteries.
  - **Ascending aorta** (three branches) – brachiocephalic, left common carotid, and subclavian arteries.



External Heart: Vessels that Supply/Drain the Heart (Anterior View)

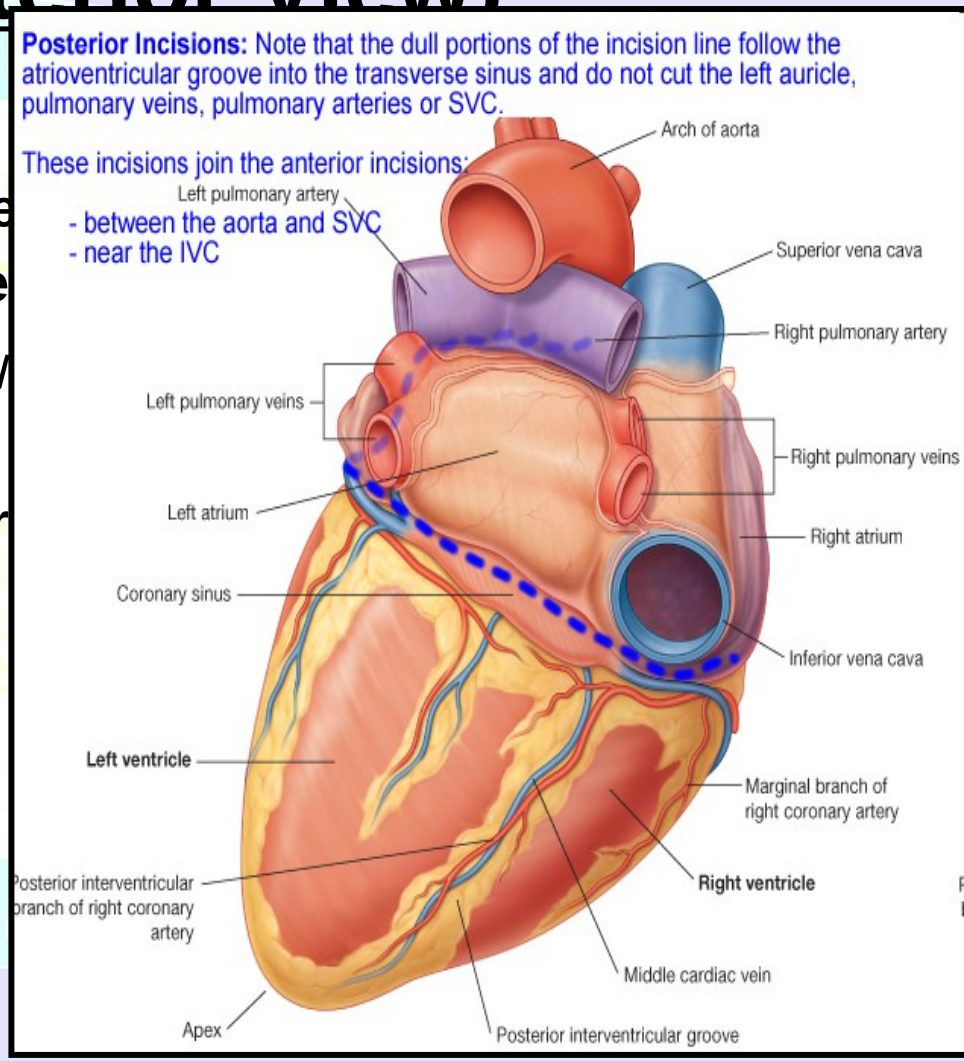
- **Arteries** – right and left coronary (in atrioventricular groove), marginal, circumflex, and anterior interventricular arteries.

- **Veins** – small cardiac, anterior cardiac, and great cardiac veins.



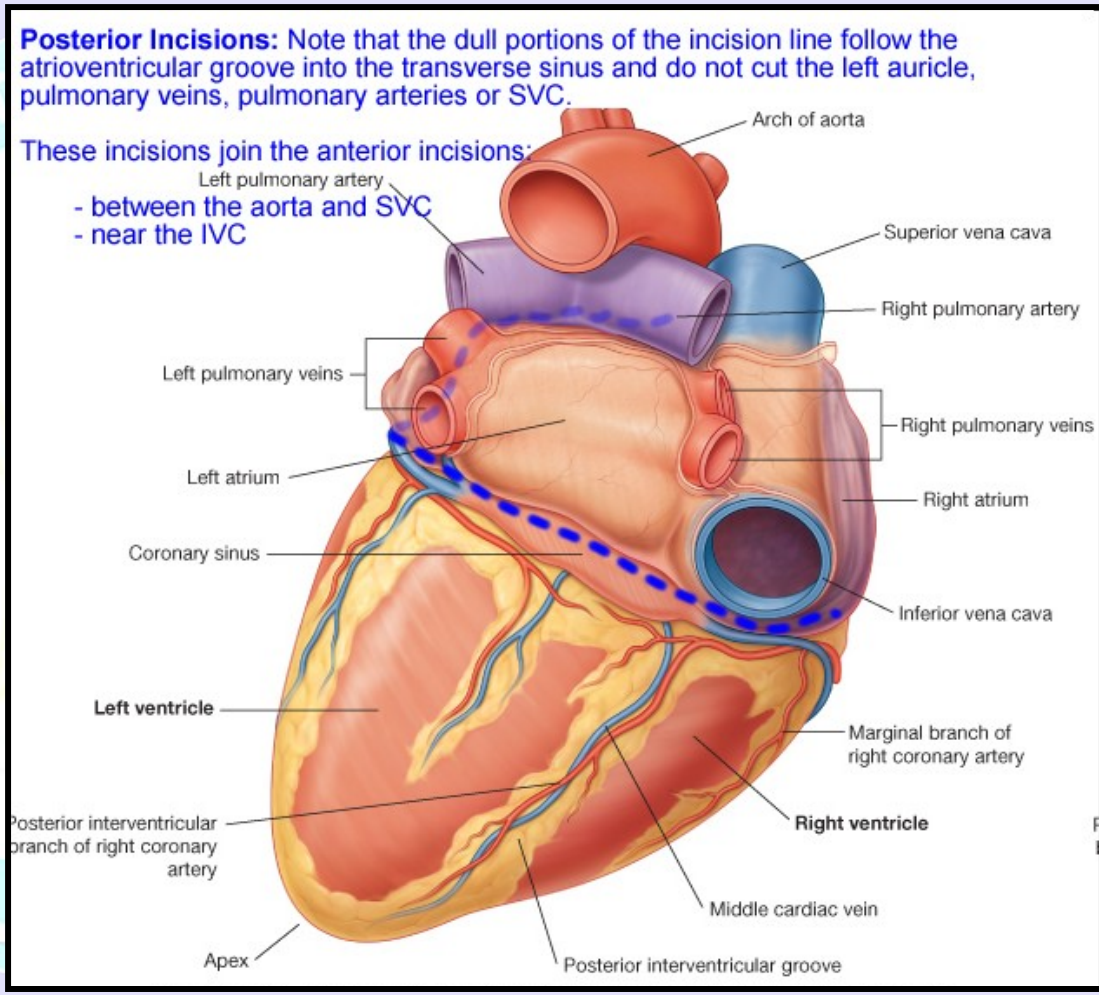
External Heart: Major Vessels of the Heart (Posterior View)

- **Vessels** returning **blood** to
  - **Right and left pulmonary ve**
  - **Superior and inferior venae**
- Vessels conveying blood aw
  - **Aorta**
  - **Right and left pulmonary ar**



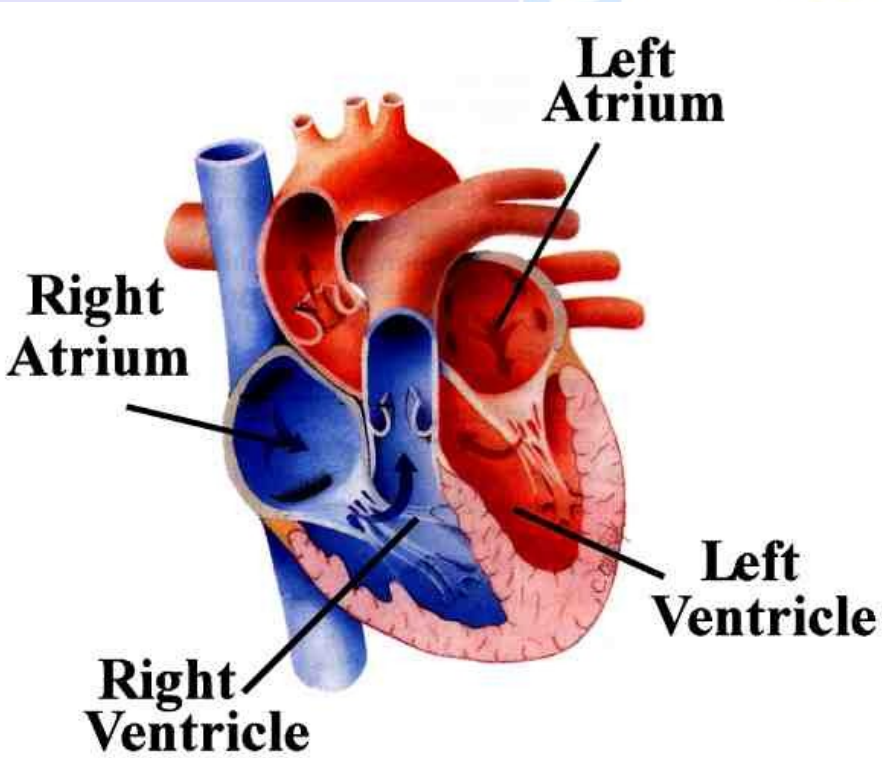
External Heart: Vessels that Supply/Drain the Heart (Posterior View)

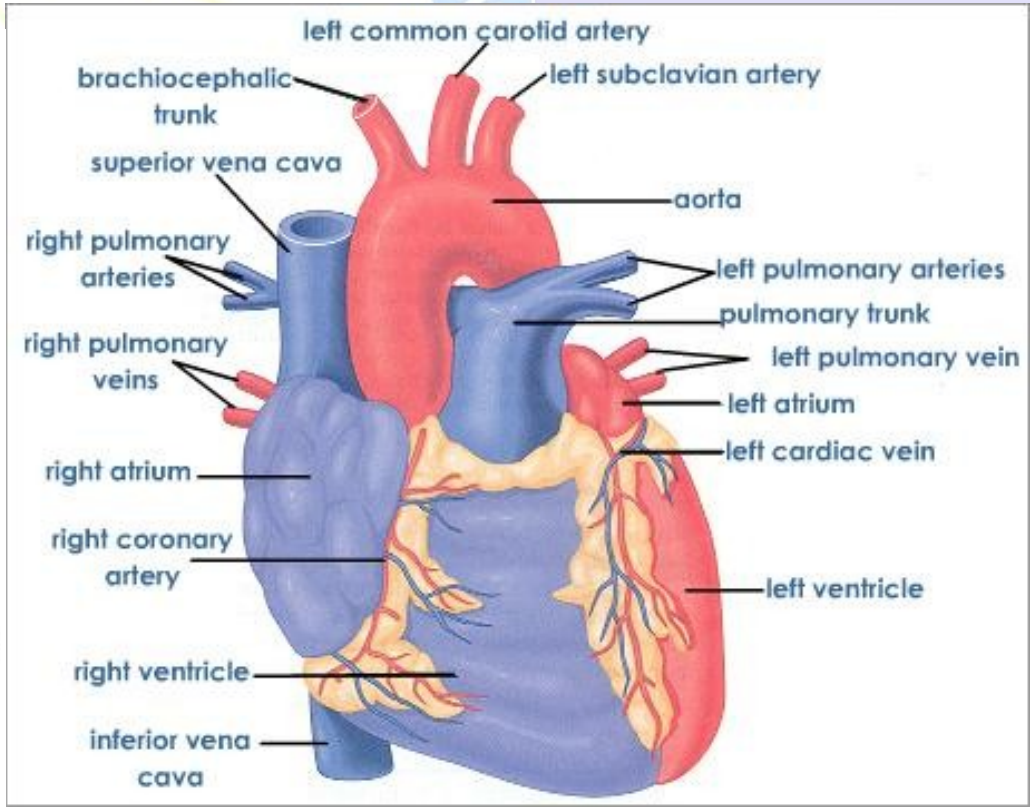
- **Arteries –** right coronary artery (in atrioventricular groove) and the posterior interventricular artery (in interventricular groove)
- **Veins** – great cardiac vein, posterior vein to left ventricle, coronary sinus, and middle cardiac vein



Atria of the Heart

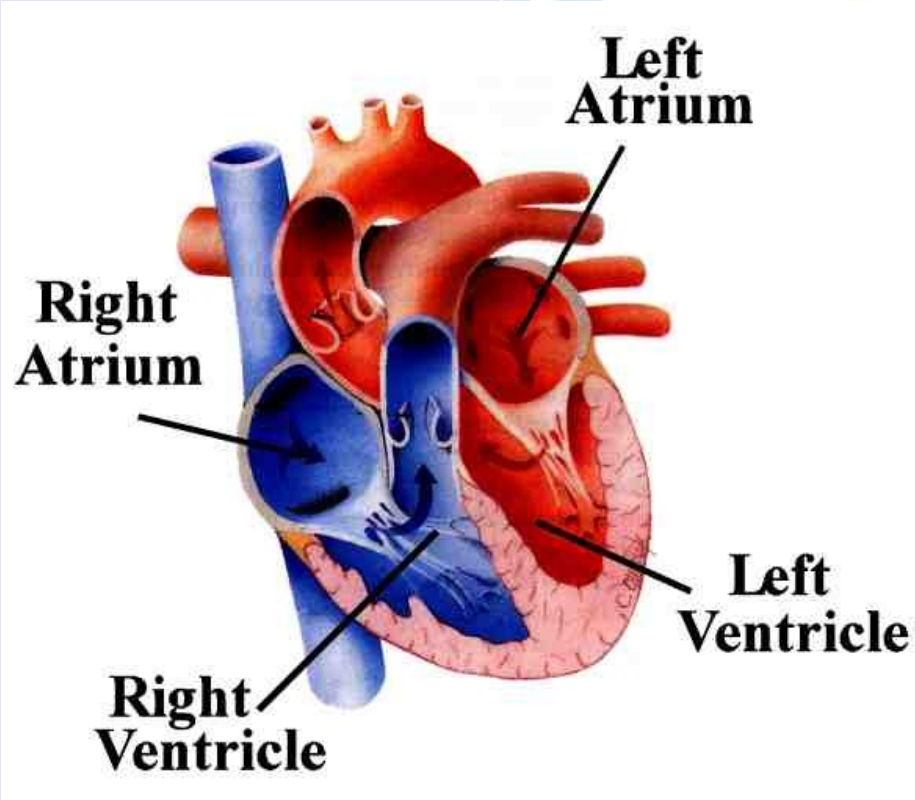
- Atria are **Receiving chambers** of the heart.
- Each atrium has a protruding **auricle.**
- **Pectinate muscles** mark atrial walls
- Blood enters right atria from **superior** and **inferior venae cavae** and **coronary sinus.**
- Blood enters left atria from **pulmonary veins.**

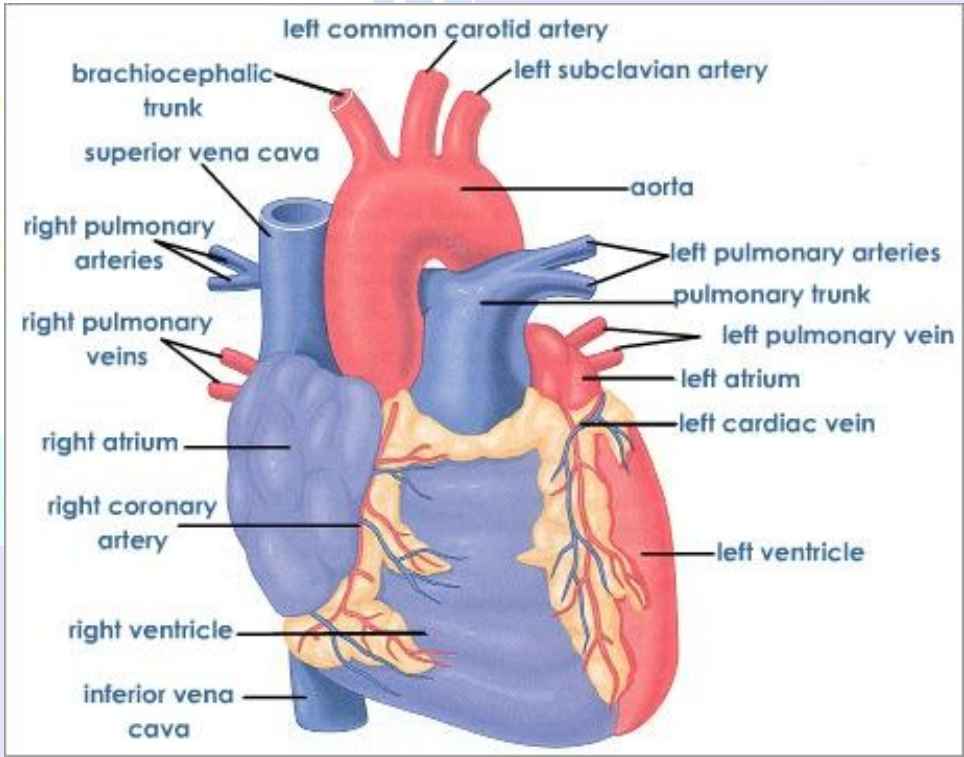




Ventricles of the Heart

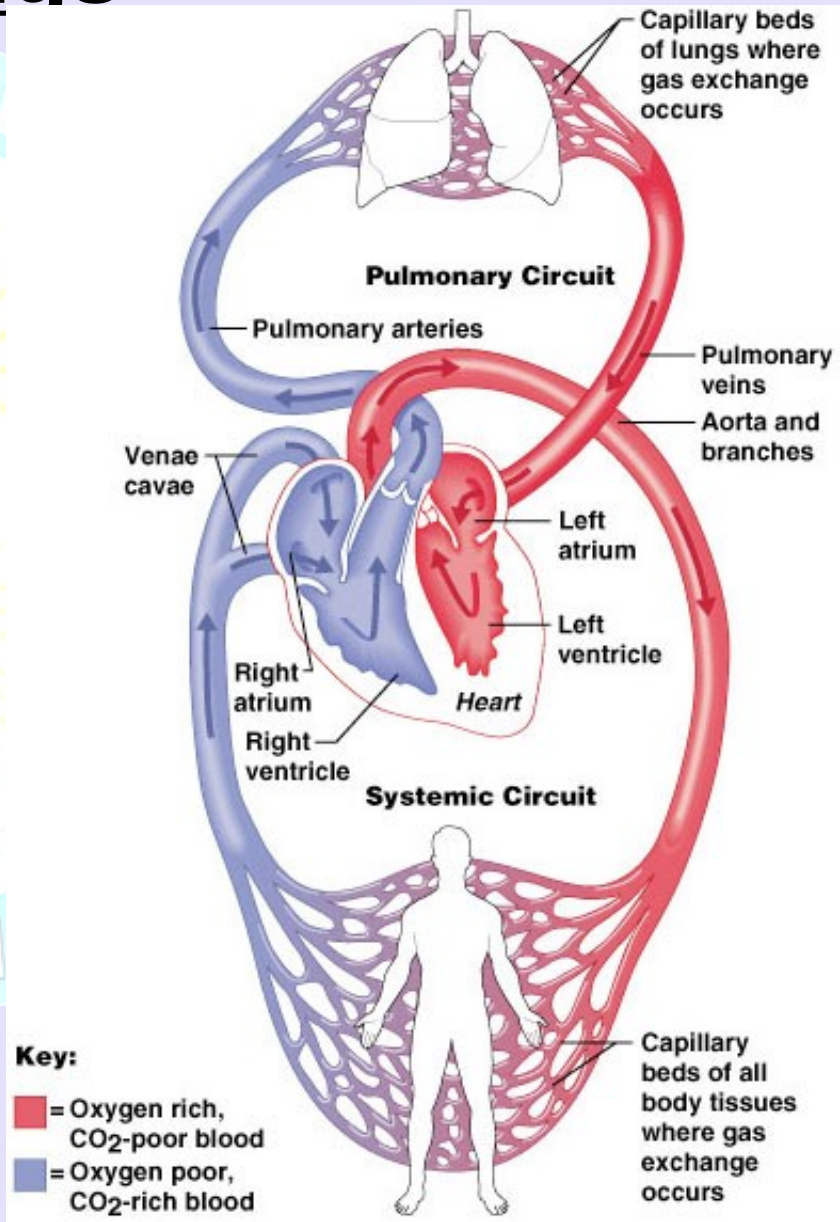
- Ventricles are the discharging chambers of the heart
- Papillary muscles and trabeculae carneae muscles mark ventricular walls
- Right ventricle pumps blood into the pulmonary trunk
- Left ventricle pumps blood into the aorta





Pathway of Blood Through the Heart and Lungs

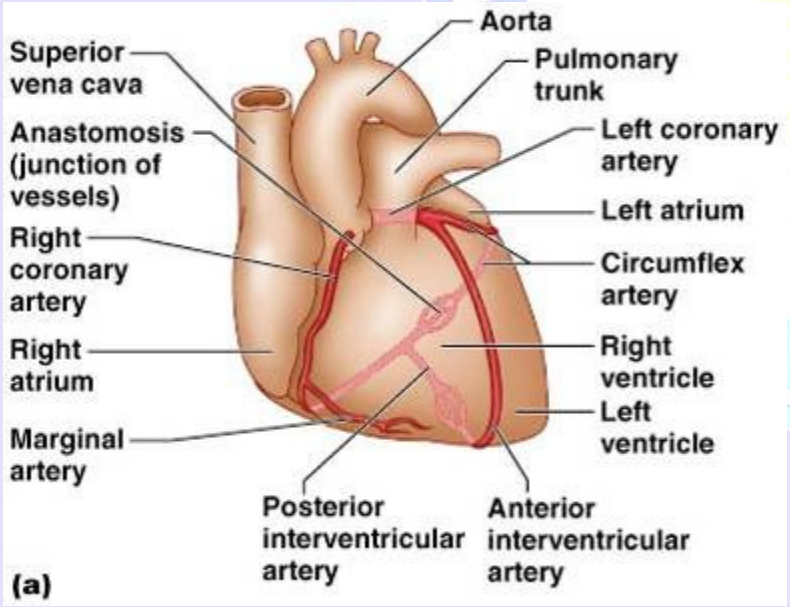
- Right atrium → tricuspid valve → right ventricle.
- Right ventricle → pulmonary semilunar valve → pulmonary arteries → lungs.
- Lungs → pulmonary veins → left atrium.
- Left atrium → bicuspid valve → left ventricle.
- Left ventricle → aortic semilunar valve → aorta.
- Aorta → systemic circulation.



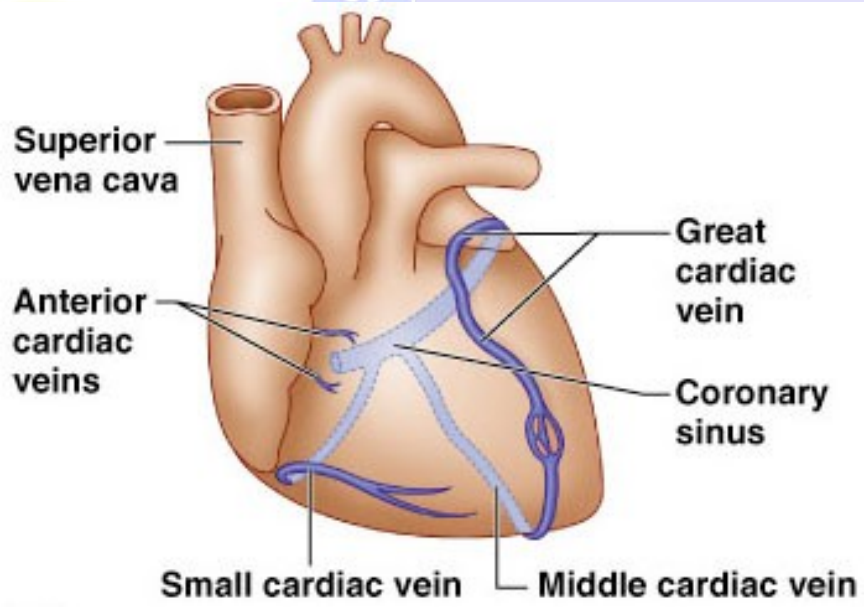
Coronary Circulation

- Coronary circulation is the functional blood supply to the heart muscle itself
- Collateral routes ensure blood delivery to heart even if major vessels are occluded

Arterial Supply

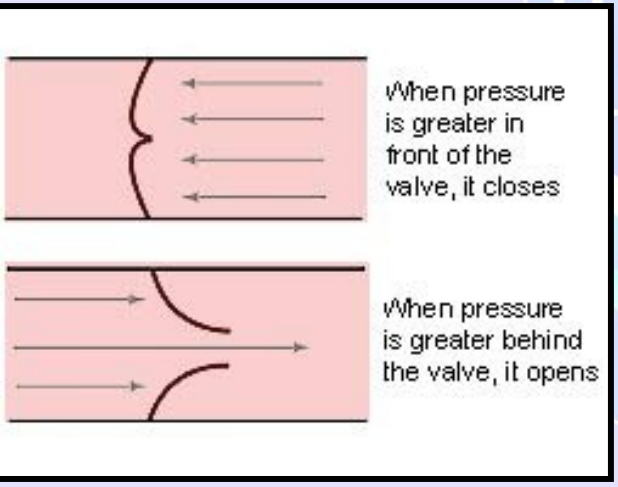


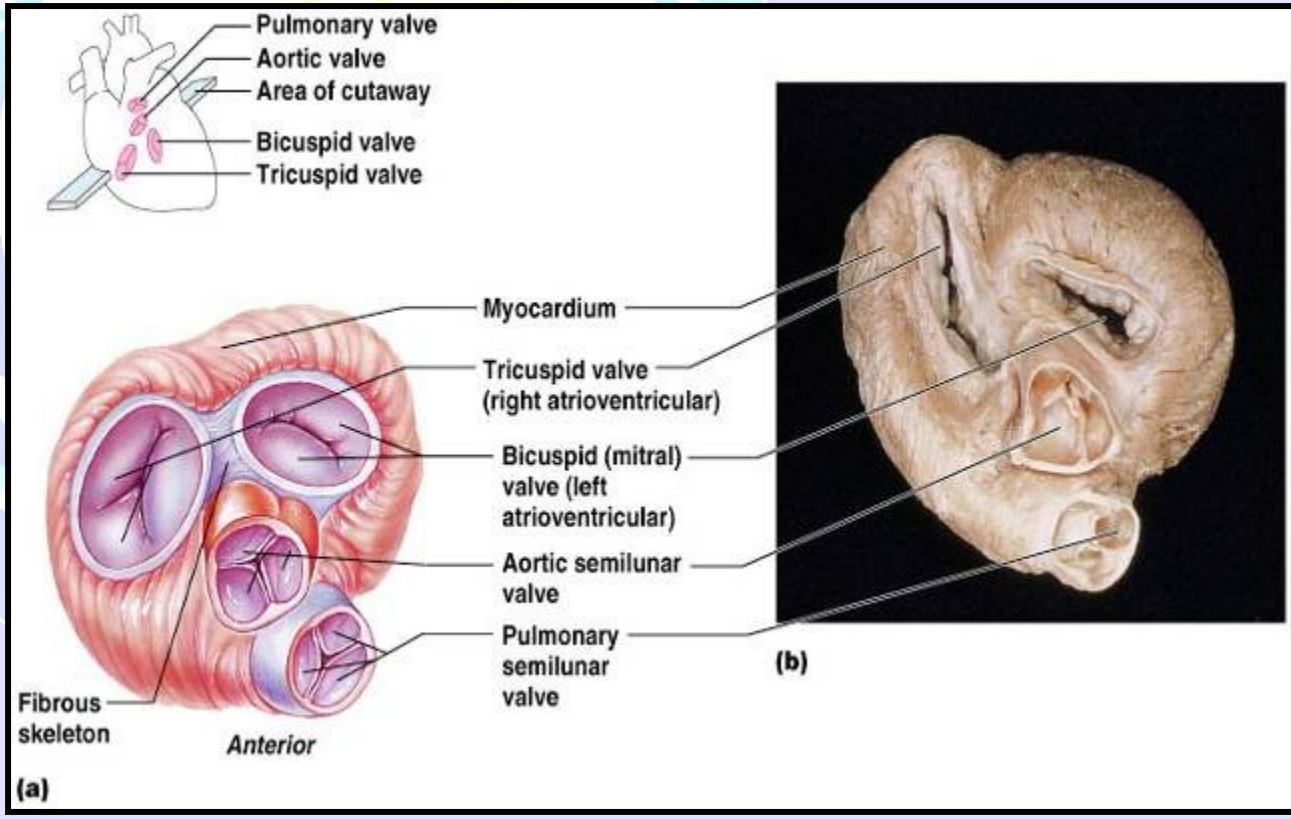
Venous Supply



Heart Valves

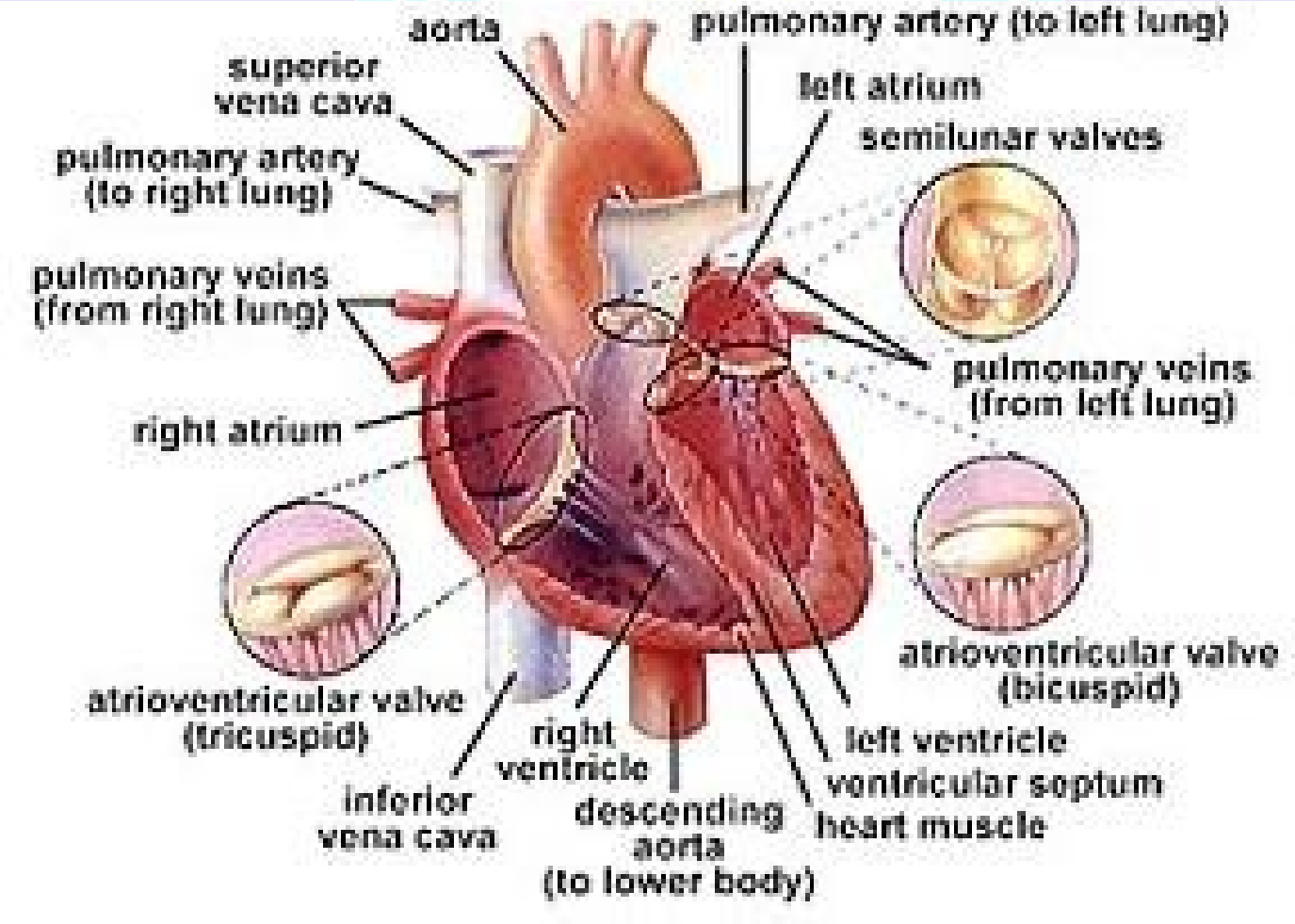
- Ensure unidirectional blood flow through heart.
- Atrioventricular (AV) valves lie between atria and ventricles.
- AV valves prevent backflow into atria when ventricles contract.
- Chordae tendineae anchor AV valves to papillary muscles.





Heart Valves

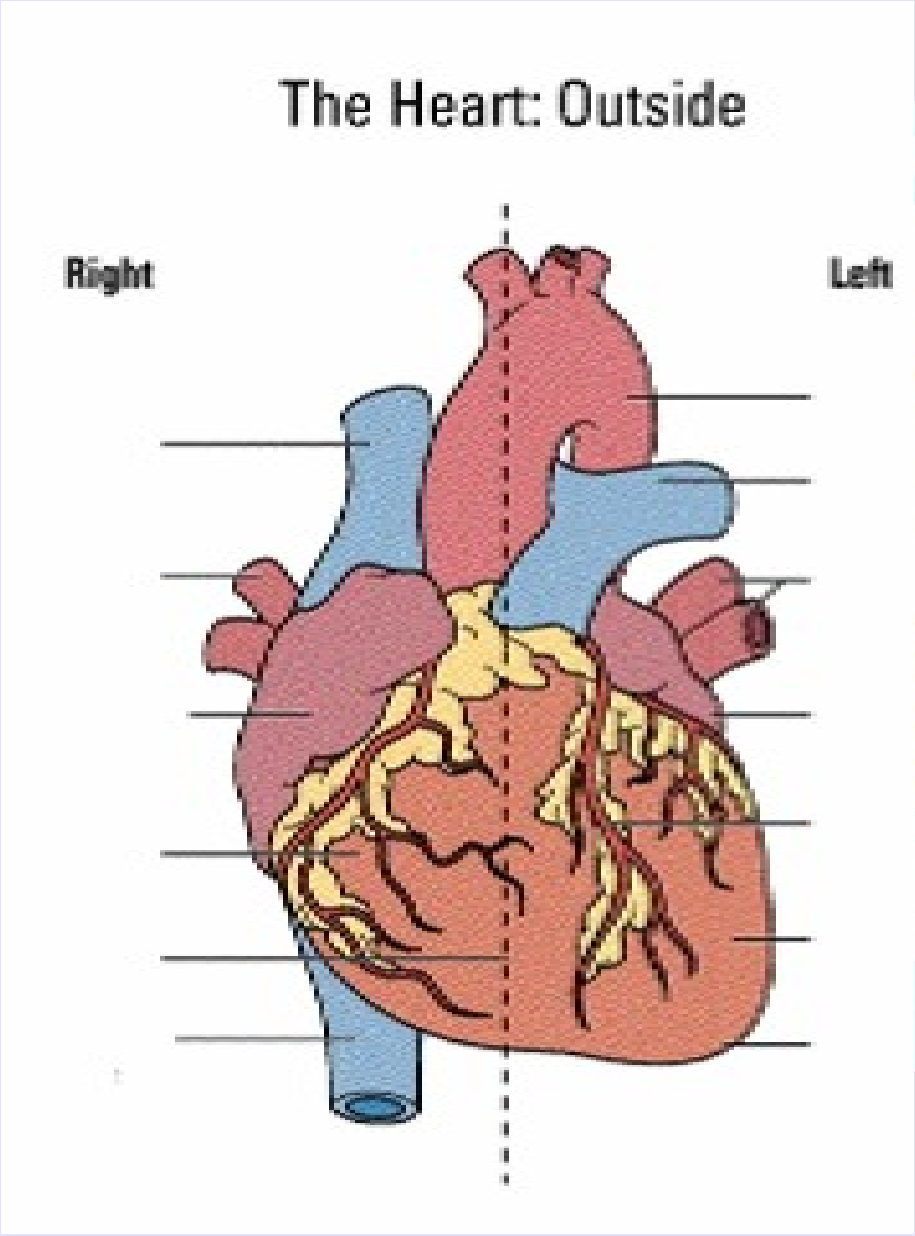
- **Aortic semilunar valve** lies between left ventricle and aorta.
- **Pulmonary semilunar valve** lies between right ventricle and pulmonary trunk.
- **Semilunar valves** prevent backflow of blood into ventricles.



References.

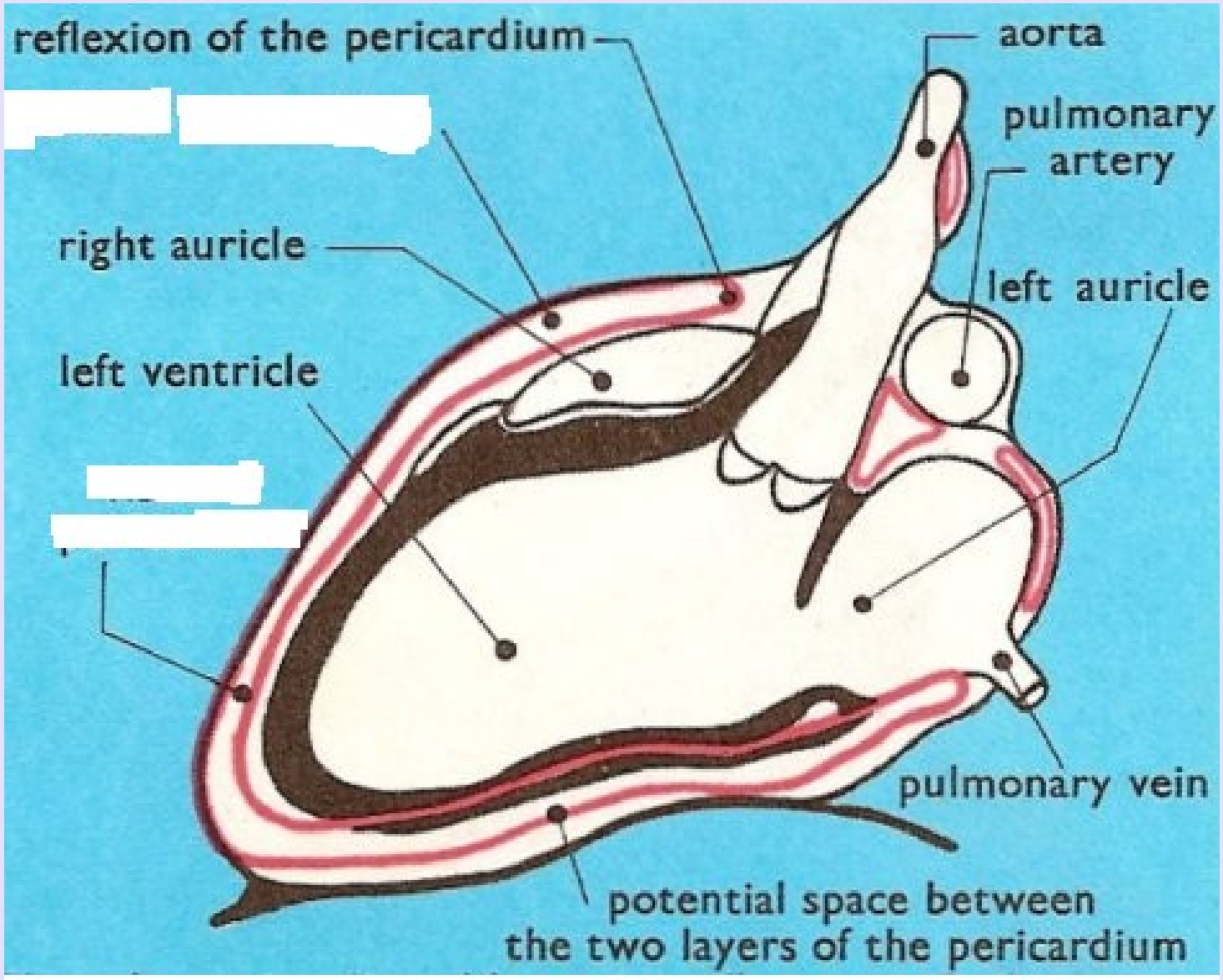
- Last's Anatomy. Regional and Applied.
- Wikipedia, the free encyclopedia

Self Assessment



- Label the diagram.

Identify the two pericardial layers.





THE END !!!!!